

OFFICIAL AGROSYNC TECHNICAL SPECIFICATION

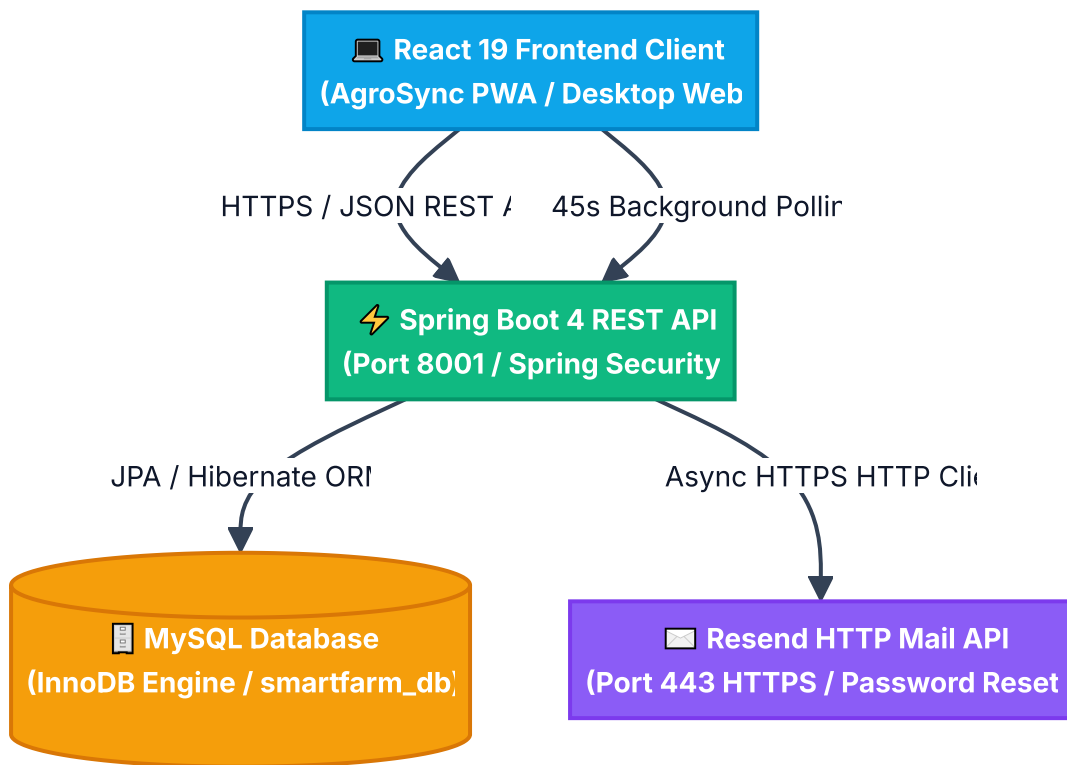
System Design and Architecture Document

AgroSync Farm Management System

Version: 4.0**SDLC Stage:** Technical Architecture & **Status:** Approved
Production Blueprint

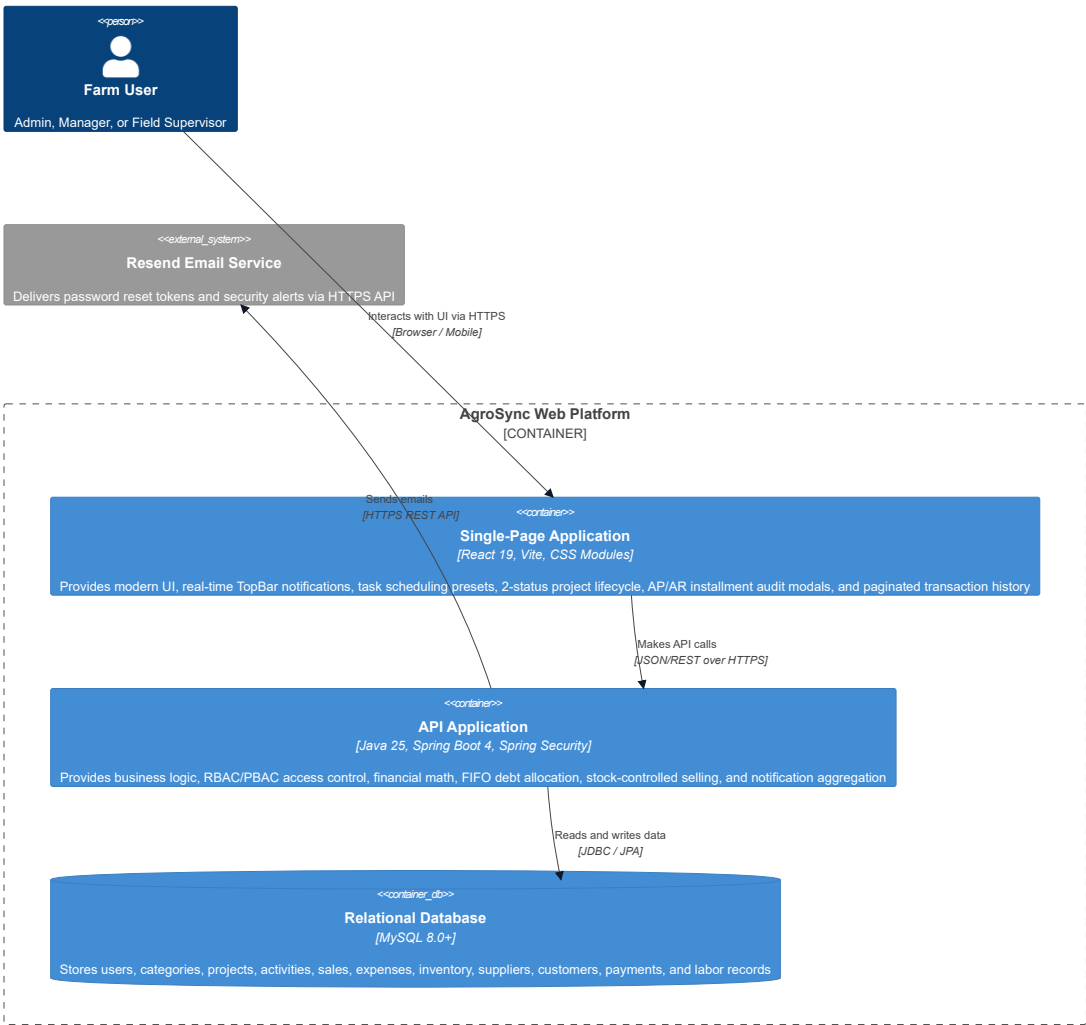
1. Executive Summary & Architectural Overview

AgroSync is built on a decoupled, cloud-native architecture combining a high-performance **Spring Boot 4 (Java 25)** micro-monolith backend with a reactive **React 19 (Vite)** single-page web client and MySQL relational database.

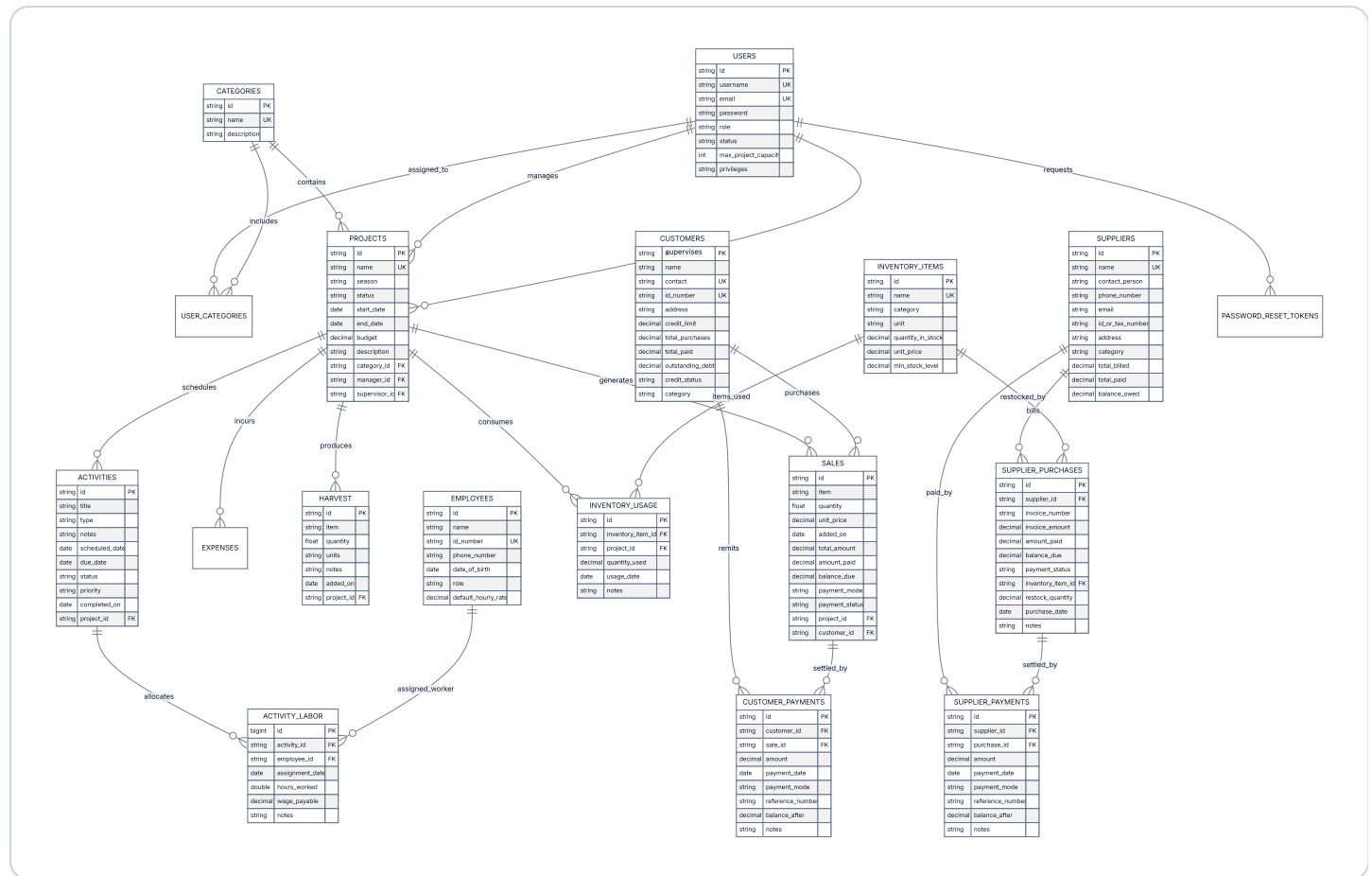


2. C4 Container Diagram

Container Diagram for AgroSync System



3. Comprehensive Entity-Relationship Data Model (ERD)



4. Financial & Business Logic Engine

4.1 Cash Received vs Booked Revenue

The financial engine computes ground-truth liquid cash vs total booked revenue:

$$\text{Total Booked Sales} = \sum_{s \in \text{Sales}} s.\text{total_amount}$$

$$\text{Pending Customer Debt (AR)} = \sum_{c \in \text{Customers}} c.\text{outstanding_debt}$$

$$\text{Cash Received (In Account)} = \text{Total Booked Sales} - \text{Pending Customer Debt}$$

4.2 Accounts Receivable (AR) & Credit Rules

For any customer C :

$$C.\text{outstanding_debt} = C.\text{total_purchases} - C.\text{total_paid}$$

$$\text{Credit Status} = \begin{cases} \text{CLEAR}, & \text{if } C.\text{outstanding_debt} = 0 \\ \text{BLOCKED}, & \text{if } C.\text{credit_limit} > 0 \text{ and } C.\text{outstanding_debt} > C.\text{credit_limit} \\ \text{HAS_DEBT}, & \text{otherwise} \end{cases}$$

4.3 Accounts Payable (AP) & Supplier Liabilities

For any supplier S :

$$S.\text{balance_owed} = S.\text{total_billed} - S.\text{total_paid}$$

4.4 Stock-Controlled Selling Validation

When a sale request R is initiated for crop item I in project P :

$$\text{Total Harvested} = \sum_{h \in P, \text{Harvest}, h.I = I} h.\text{quantity}$$

$$\text{Total Sold} = \sum_{s \in P, \text{Sales}, s.I = I} s.\text{quantity}$$

$$\text{Available Stock} = \max(0, \text{Total Harvested} - \text{Total Sold})$$

$$\text{Validation Rule: If } R.\text{quantity} > \text{Available Stock} \implies \text{Reject Sale (HTTP 400)}$$

4.5 FIFO Installment Payment Settlement Algorithm

When an unallocated general debt payment A is submitted for customer C (or supplier S):

```

1. Fetch all unpaid/partial sales of customer sorted by added_on ASC (Oldest First).
2. Remaining Payment R = A
3. FOR EACH sale S IN unpaid_sales:
    IF R ≤ 0 THEN BREAK
    Available To Settle = S.balance_due
    Payment Allocated P = MIN(R, Available To Settle)
    S.amount_paid += P
    S.balance_due -= P
    IF S.balance_due = 0 THEN S.payment_status = "PAID_IN_FULL"
    ELSE S.payment_status = "PARTIAL_PAYMENT"
    Save S
    Log CustomerPayment Record (amount = P, balance_after = S.balance_due)
    R -= P
4. C.total_paid += A
5. C.outstanding_debt = C.total_purchases - C.total_paid
6. Update customer credit status (CLEAR / HAS_DEBT / BLOCKED)

```

5. Complete REST API Catalog

Module	Method	Endpoint	Access Role	Description
Auth	POST	/auth/register	Public	Registers users (Auto-promotes 1st user to ADMIN).
	POST	/auth/login	Public	Authenticates credentials and returns user payload.
	POST	/auth/forgot-password	Public	Generates 15-min token and sends email via Resend API.
	POST	/auth/reset-password	Public	Validates token and resets user password.
Projects	POST	/projects/create	Admin / Manager	Creates new project with status 'active' or 'completed'.
	GET	/projects/{category_id}/{category}	All	Returns paginated projects scoped by user portfolio.
	GET	/projects/{projectId}	All	Returns full project details, tabs, and financial records.
	PUT	/projects/{id}	Admin / Manager	Updates project configuration, season, dates, and budget.
	PATCH	/projects/{id}/status	Admin / Manager / Sup	Toggles project status between 'active' and 'completed'.
Activities	POST	/activities/record	Admin / Mgr / Sup	Creates activity with agronomic preset or custom dates.
	PATCH	/activities/{id}/status	Admin / Mgr / Sup	Updates activity lifecycle status ('SCHEDULED', 'COMPLETED', etc.).
	POST	/activities/{id}/labor	Admin / Mgr / Sup	Assigns verified employee labor and logs wage payables.
Harvest	POST	/harvest/record	Admin / Mgr / Sup	Logs crop or livestock yield volume.
Sales	POST	/sales/record	Admin / Mgr / Sup	Records farm-gate sale with stock verification & down-payment.
Customers	GET	/customers	All	Lists all customers with cumulative AR balances.
	GET	/customers/{id}	All	Returns customer profile and Accounts Receivable ledger.

Module	Method	Endpoint	Access Role	Description
	`GET`	`/customers/{id}/sales`	All	Returns paginated sales history for this customer.
	`GET`	`/customers/sales/{saleId}/payments`	All	Returns chronological installment audit trail for a sale.
	`POST`	`/customers/{id}/payments`	Admin / Mgr / Sup	Records installment payment (direct sale or FIFO).
Suppliers	`GET`	`/suppliers`	Admin / Manager	Lists suppliers with cumulative Accounts Payable balances.
	`GET`	`/suppliers/{id}`	Admin / Manager	Returns supplier profile and Accounts Payable ledger.
	`GET`	`/suppliers/{id}/purchases`	Admin / Manager	Returns purchase invoice and delivery audit trail.
	`GET`	`/suppliers/purchases/{purchaseId}/payments`	Admin / Manager	Returns chronological voucher payment audit trail.
	`POST`	`/suppliers/purchases`	Admin / Manager	Records purchase invoice with optional auto-restock.
	`POST`	`/suppliers/{id}/payments`	Admin / Manager	Records debt payment voucher (direct invoice or FIFO).
Inventory	`GET`	`/inventory`	All	Lists warehouse items with quantities and unit prices.
	`POST`	`/inventory/{id}/use`	Admin / Mgr / Sup	Deducts supplies consumed directly by a farm project.
Dashboard	`GET`	`/dashboard/summary`	Admin / Manager	Aggregates KPIs, Sector Donut, Project Donut, and charts.
	`GET`	`/dashboard/transactions`	Admin / Manager	Returns paginated unified transaction feed (5 per page).
	`GET`	`/supervisor/projects/{userId}`	Supervisor	Returns scoped supervisor field workspace.
Notifs	`GET`	`/notifications`	All	Aggregates real-time alerts across 4 categories.

6. Security Architecture & Exception Handling

6.1 Stateless Header Authentication

Requests transmit `X-User-Id` and `X-User-RoLe` .

Spring Security filter chain validates credentials against database records on every request.

Passwords stored as BCrypt hashes with salt.

6.2 Serialization Protection

Bidirectional JPA relationships (`Sale` $\rightarrow$ `Project` , `Activity` $\rightarrow$ `Project` , `Harvest` $\rightarrow$ `Project` , `InventoryUsage` $\rightarrow$ `Project` , `ActivityLaborAssignment` $\rightarrow$ `Activity`) are annotated with `@JsonIgnore` on both fields and getters to prevent Jackson circular recursion and `StackOverflow` errors.

6.3 Global Error Handling

`GlobalExceptionHandler` intercepts exceptions and formats unified HTTP JSON responses:

```
{
  "body": null,
  "message": "Descriptive error message",
  "success": false,
  "time": "2026-09-11T08:45:00Z"
}
```